



NSW Education Standards Authority

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Centre Number

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Student Number

2025 HIGHER SCHOOL CERTIFICATE EXAMINATION

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- Write your Centre Number and Student Number at the top of this page

Total marks: 100

Section I – 20 marks (pages 2–11)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 13–36)

- Attempt Questions 21–34
- Allow about 2 hours and 25 minutes for this section

Section I

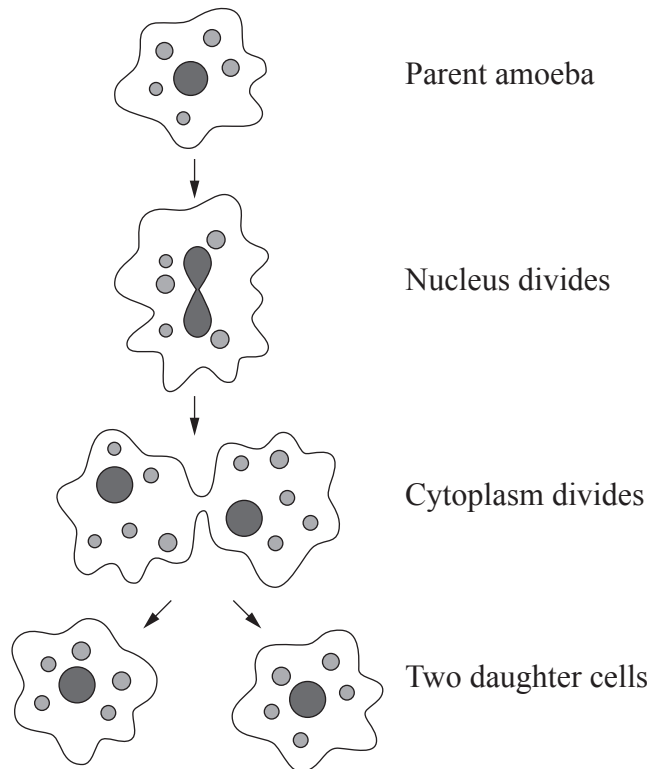
20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

1 Amoeba reproduce by the process shown.



© Achiichiii / Dreamstime.com

Which of the following is a characteristic of the daughter cells?

- A. Genetically different to parent amoeba
- B. Genetically identical to parent amoeba
- C. More organelles due to the cell division
- D. Fewer chromosomes than the parent amoeba

2 What type of mutagen is UV light?

- A. Biochemical
- B. Biological
- C. Chemical
- D. Electromagnetic

3 Which row in the table correctly identifies the role of phagocytes and lymphocytes?

	<i>Phagocytes</i>	<i>Lymphocytes</i>
A.	Engulf bacteria	Produce antibodies
B.	Engulf bacteria	Produce antigens
C.	Produce antibodies	Engulf bacteria
D.	Produce antigens	Produce antibodies

4 A storm randomly kills 80% of the frog population on an island. The allele frequencies in the frog population after the storm are notably different to those of the population before the storm.

What is the process that has led to the change in allele frequencies?

- A. Gene flow
- B. Genetic drift
- C. Natural selection
- D. Survival of the fittest

5 An infectious disease is spread through direct contact between hosts.

Under which conditions will this disease spread most rapidly?

- A. Low population density with infected individuals rarely dying from the disease
- B. High population density with infected individuals rarely dying from the disease
- C. Low population density with infected individuals quickly dying from the disease
- D. High population density with infected individuals quickly dying from the disease

- 6 Rabies is a viral disease spread by infected animals. If bitten by an infected animal, a person can be treated by receiving an injection of antibodies.

What type of immunity will this person have following the injection?

- A. Innate active
 - B. Natural passive
 - C. Acquired active
 - D. Acquired passive
- 7 An animal's body temperature and the air temperature of the animal's environment were measured every 4 hours, and the following data were recorded.

<i>Time</i>	<i>Body temperature (°C)</i>	<i>Air temperature (°C)</i>
4 am	41.3	19.2
8 am	41.1	18.8
12 pm	40.8	21.5
4 pm	41.4	26.4
8 pm	41.2	27.5
12 am	41.5	23.0

Based on this data, which row of the table indicates what type of animal it is and why?

	<i>Type of animal</i>	<i>Reason</i>
A.	Ectotherm	Body temperature is around 41°C and varies with the air temperature.
B.	Ectotherm	Body temperature is around 41°C and is always above the air temperature.
C.	Endotherm	Body temperature is relatively constant despite changes in air temperature.
D.	Endotherm	Body temperature is relatively constant, and air temperature is relatively constant.

- 8 When a red camellia flower is crossed with a white camellia flower, all the offspring are covered in both red and white petals.

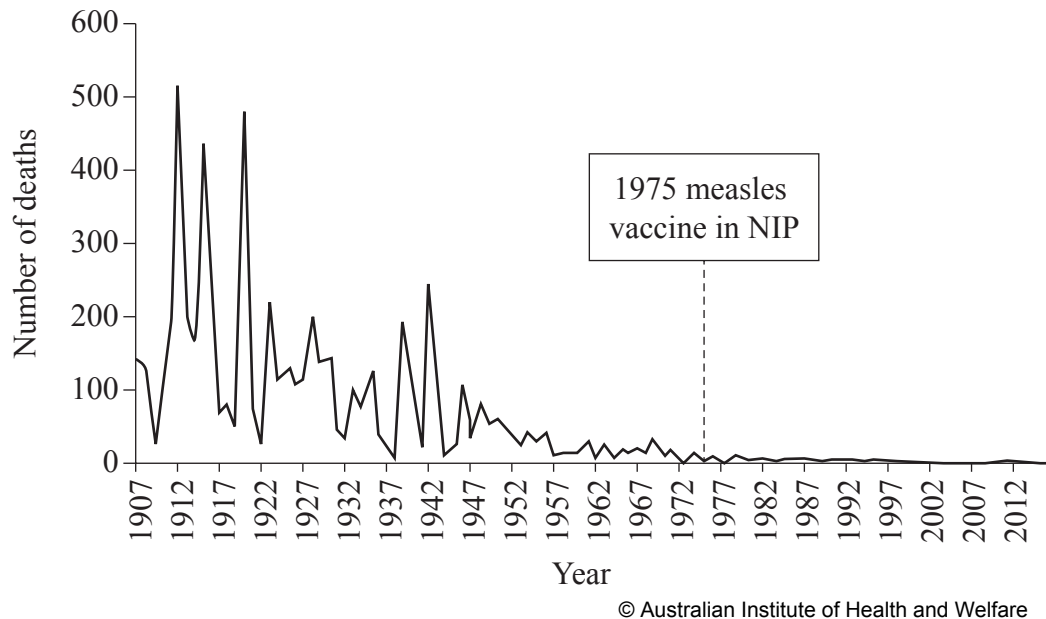
What is the reason for this occurrence?

- A. One gene is controlling multiple characteristics.
 - B. Environmental factors affect the phenotype of camellia flowers.
 - C. Alleles for both red and white colour in camellia flowers are recessive.
 - D. Petal colour in camellia flowers is controlled by a co-dominance pattern of inheritance.
- 9 Dialysis is used to assist people with loss of kidney function.

Which row of the table correctly describes the composition of blood before and after dialysis?

	<i>Blood before dialysis</i>	<i>Blood after dialysis</i>
A.	High urea and high glucose	Low urea and high glucose
B.	High urea and high glucose	High urea and low glucose
C.	Low urea and low glucose	Low urea and high glucose
D.	Low urea and high glucose	Low urea and low glucose

- 10** The graph shows the number of recorded deaths, due to measles, before and after the measles vaccine was included in the National Immunisation Program (NIP).



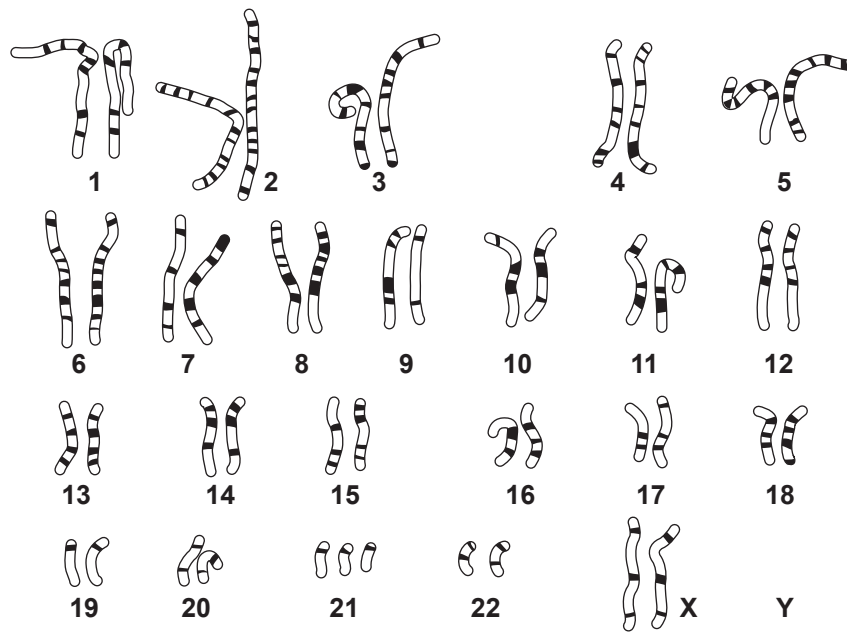
Which of the following is a trend shown in the graph?

- A. Most people have been immunised against measles since 1975.
 - B. Over the last 100 years the number of deaths has consistently fallen.
 - C. The number of measles-induced deaths has fallen to near zero since measles vaccine was added to the NIP.
 - D. The National Immunisation Program was the primary reason for the great reduction in measles-induced deaths.
- 11** In 2018, the Victorian government reported the target of vaccinating more than 95% of children below the age of five had been achieved.

What is the benefit of achieving a 95% vaccination rate for an infectious disease?

- A. Only 5% of individuals will catch the disease.
- B. Only 5% of individuals will be protected by the vaccine.
- C. This will protect the remaining 5% who are not vaccinated.
- D. The vaccinated individuals can transfer immunity to the remaining 5%.

- 12 The following represents a karyotype for an individual with Down syndrome.



© National Down Syndrome Society

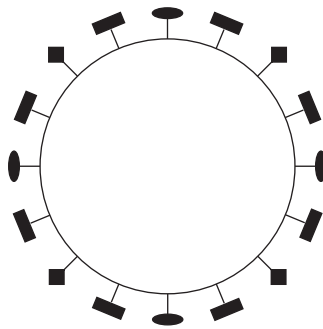
What conclusions can be drawn about the individual from this karyotype?

- A. A male with a deletion mutation
 - B. A female with a deletion mutation
 - C. A male with a chromosomal mutation
 - D. A female with a chromosomal mutation
- 13 The prevalence of a non-infectious disease has remained constant for over 10 years. A new treatment for this disease prolongs the life of people with the disease, but does not cure them.

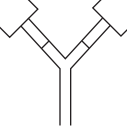
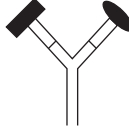
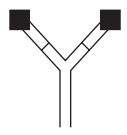
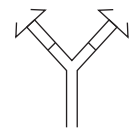
Which row in the table shows the effect of the treatment on both the incidence and prevalence of this disease?

	<i>Change in incidence</i>	<i>Change in prevalence</i>
A.	None	Increases
B.	None	Decreases
C.	Increases	None
D.	Decreases	None

- 14 A pathogen that produces an immune response is shown.



Which antibody will be produced as a response to the pathogen?

- A. 
- B. 
- C. 
- D. 

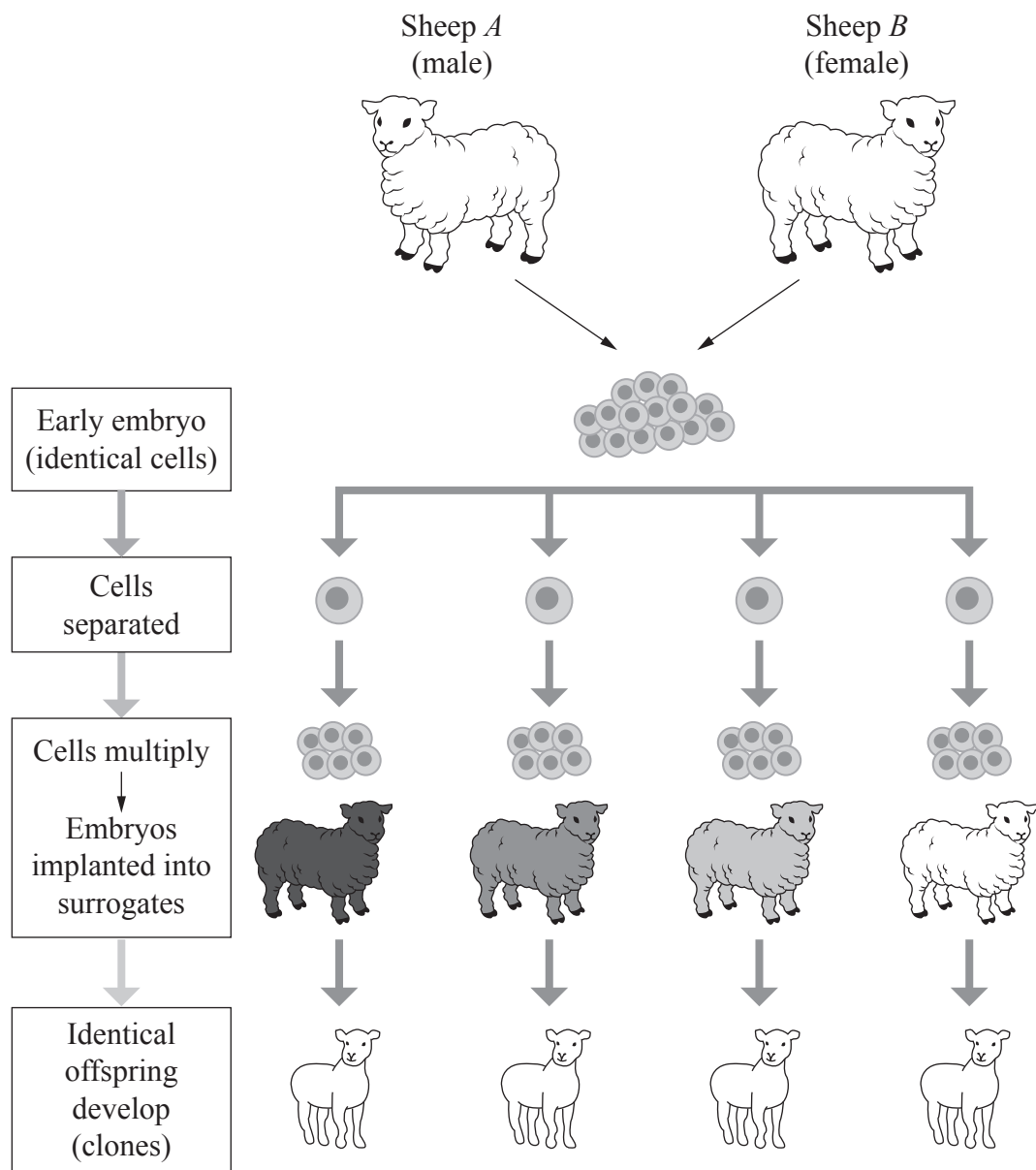
- 15 Cochlear implants assist with hearing. The following are five steps involved in the process.

1. The sound signal is turned into electrical impulses.
2. The implant sends electrical impulses to the electrodes in the cochlea.
3. Sounds are picked up by the microphone.
4. The auditory nerve picks up and sends electrical impulses to the brain.
5. The electrical impulses are transmitted across the skin to the implant.

Which is the correct order for this process?

- A. 1, 4, 3, 2, 5
- B. 1, 3, 2, 4, 5
- C. 3, 1, 5, 2, 4
- D. 3, 1, 2, 5, 4

- 16 The diagram shows a process that was used to make multiple clones in sheep.

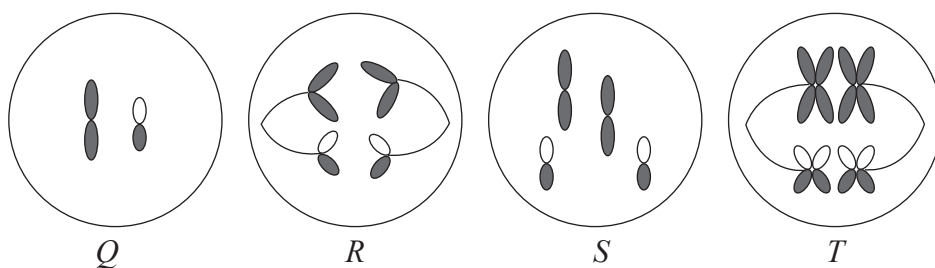


© VCE BioNinja

The identical offspring are clones of

- A. each other.
- B. only sheep *B*.
- C. the surrogates.
- D. both sheep *A* and *B*.

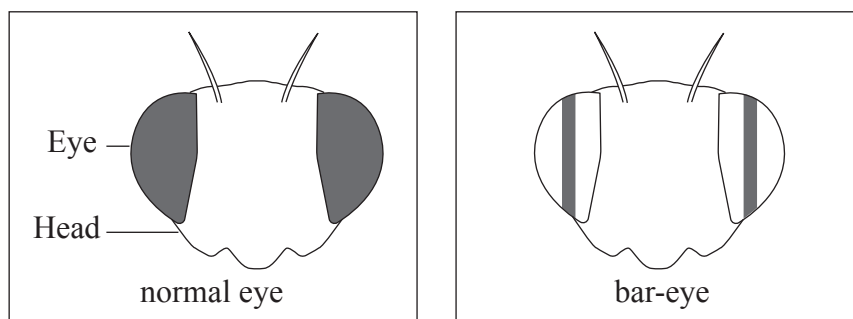
- 17 Some phases of cell division in an organism are shown.



What is the correct sequence of the process?

- A. Q, S, R, T
 B. Q, R, S, T
 C. S, T, R, Q
 D. S, R, T, Q
- 18 The fruit fly, *Drosophila melanogaster*, normally has large round eyes. A mutation of the eye shape gene, found on the X-chromosome, causes the eye to be a narrow slit (bar-eye), which is a dominant allele for eye shape.

Offspring were produced when a bar-eyed male was crossed with a normal female. Males are XY and females are XX.



Which row in the table shows the correct percentage of male and female offspring produced with bar-eye?

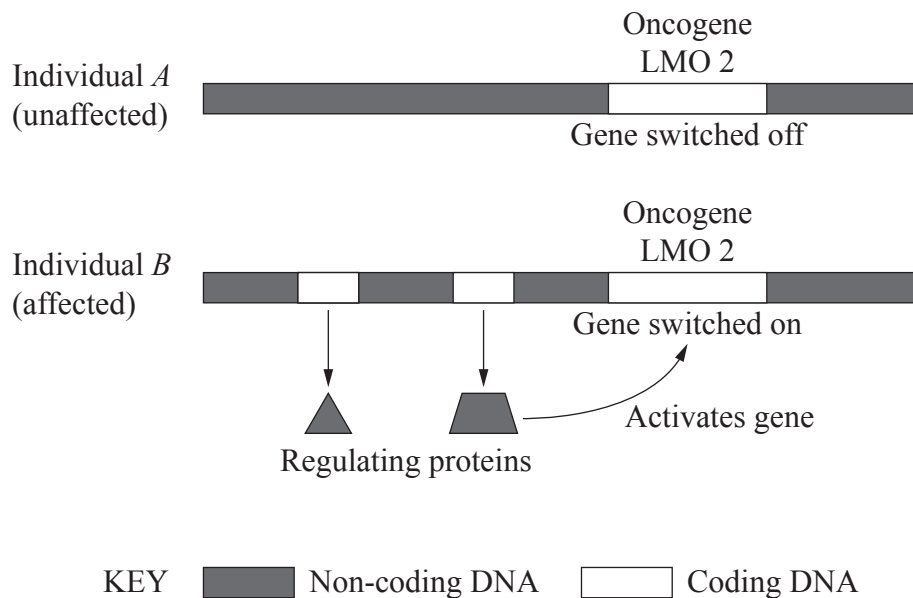
<i>Percentage of offspring with bar-eyes</i>		
	<i>Males</i>	<i>Females</i>
A.	100	100
B.	0	100
C.	100	0
D.	50	0

- 19 The grasshopper has an average genome size of 120 million base pairs. 42% of these base pairs contain cytosine and guanine.

What is the number of DNA nucleotides with a thymine base found in each cell?

- A. 25.2 million
B. 34.8 million
C. 69.6 million
D. 120 million
- 20 Researchers studying T-cell acute lymphoblastic leukaemia (T-ALL) examined a section of DNA in individuals *A* and *B*.

In individual *B*, they found proteins that can regulate the expression of oncogene LMO 2. The following diagram represents the sections of DNA from the two individuals.



Based on this information, which of the following statements is the best explanation of the cause of T-ALL?

- A. T-ALL is caused by an inactive Oncogene LMO 2.
B. The regulating proteins are the direct cause of T-ALL.
C. The regulating proteins have no effect as they are made on non-coding DNA.
D. The regulating proteins can activate the Oncogene LMO 2, causing the condition.

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Centre Number

Biology

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Student Number

Section II Answer Booklet

80 marks

Attempt Questions 21–34

Allow about 2 hours and 25 minutes for this section

Instructions

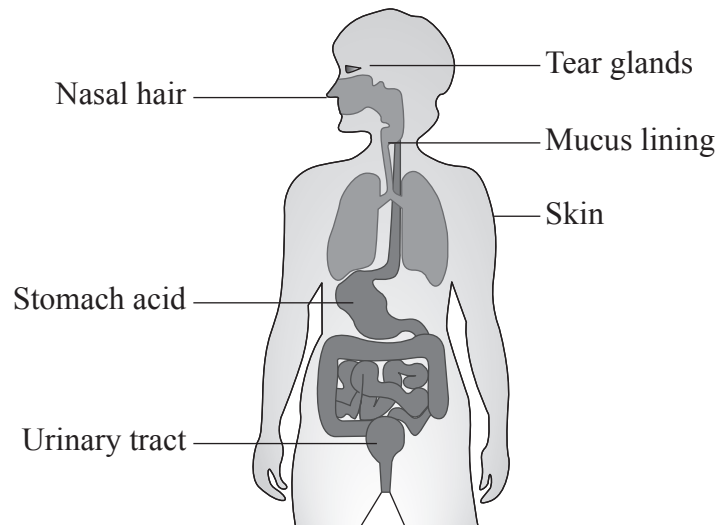
- Write your Centre Number and Student Number at the top of this page
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response
- Show all relevant working in questions involving calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering

Please turn over

Question 21 (2 marks)

The diagram shows components of the innate immune system in humans.

2



Courtesy of Dr. Marcus Judge & Dr. Ethan Kavanagh – SimpleMed

State the role of TWO components that protect against infection.

<i>Component</i>	<i>How it protects against infection</i>

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Question 22 (4 marks)

(a) Outline a process used by fungi for reproduction.

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(b) Outline an adaptation in a pathogen that facilitates transmission between hosts.

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Question 23 (3 marks)

Compare the processes of artificial insemination and artificial pollination.

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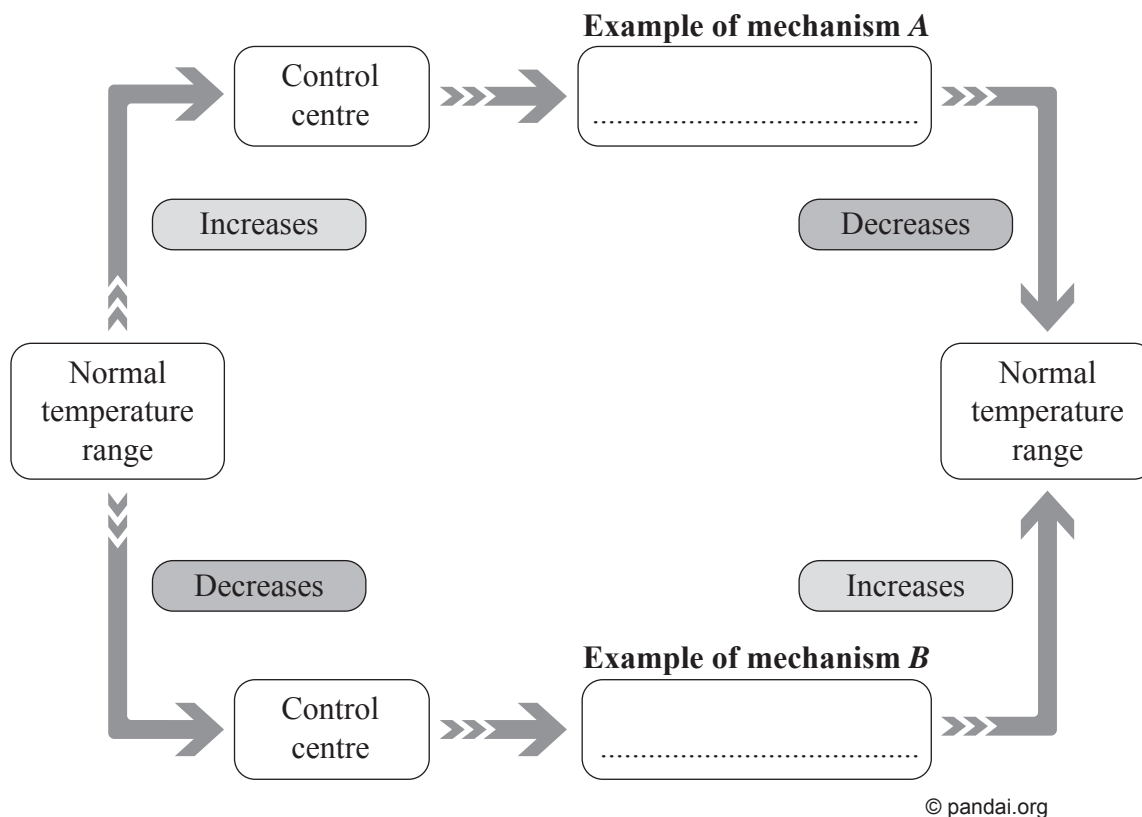
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Question 24 (4 marks)

The following flow chart represents the control of body temperature in humans.



- (a) Complete the flow chart to give an example of mechanism *A* and an example of mechanism *B*. 2
- (b) Outline how mechanism *B* maintains homeostasis. 2

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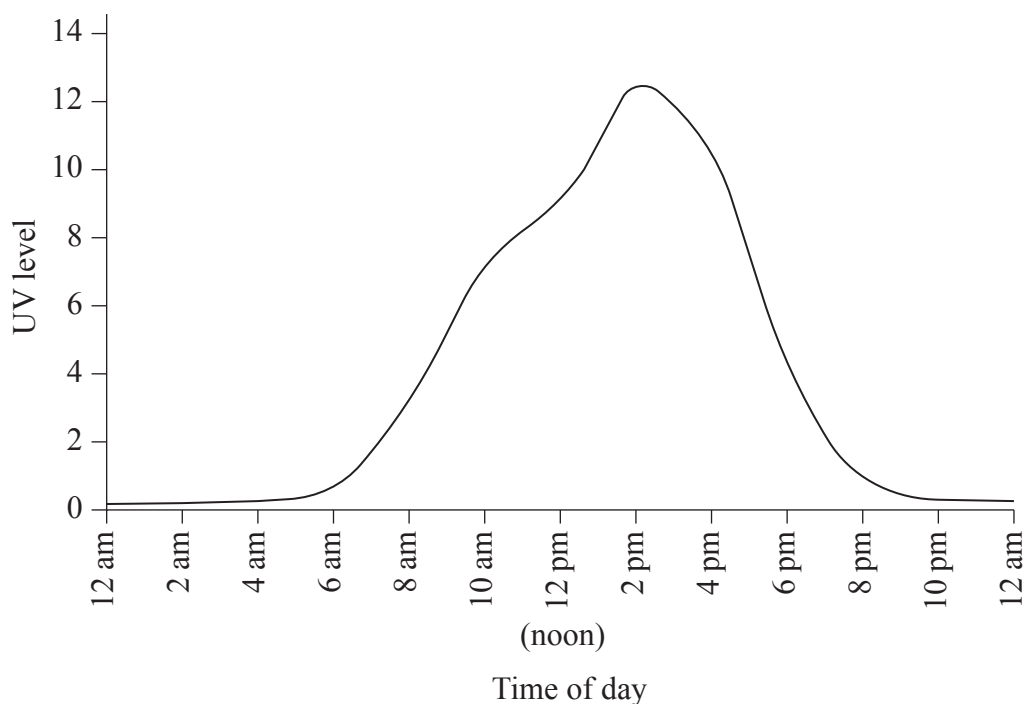
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Question 25 (3 marks)

The graph shows the changes in UV level in a single day.

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The Cancer Council suggests that sun protection is needed whenever the UV level is 3 or above. The information provided on a sunscreen product suggests that sunscreen should be used between 10 am and 4 pm.

Using the graph, evaluate the information provided on the sunscreen product with regard to the Cancer Council suggestion.

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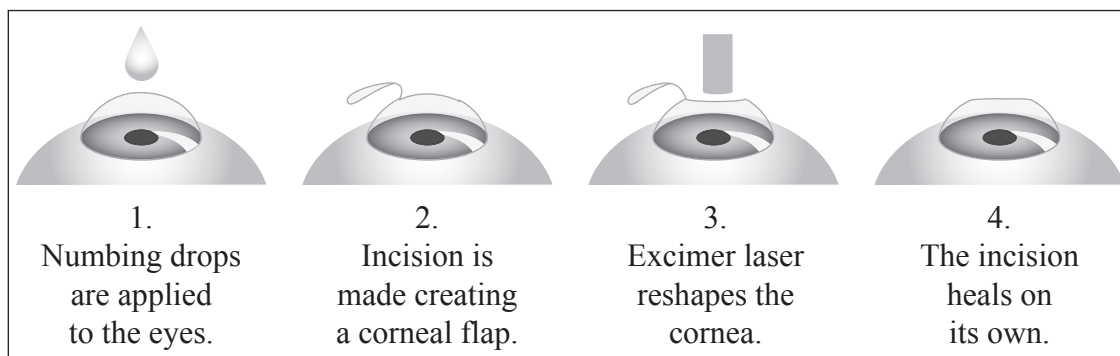
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Question 26 (4 marks)

The diagram shows the steps in LASIK (Laser-Assisted In Situ Keratomileusis) surgery.

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© Shutterstock Ireland

Compare the LASIK technology shown with ONE other technology that can be used to treat a named visual disorder.

Visual disorder:

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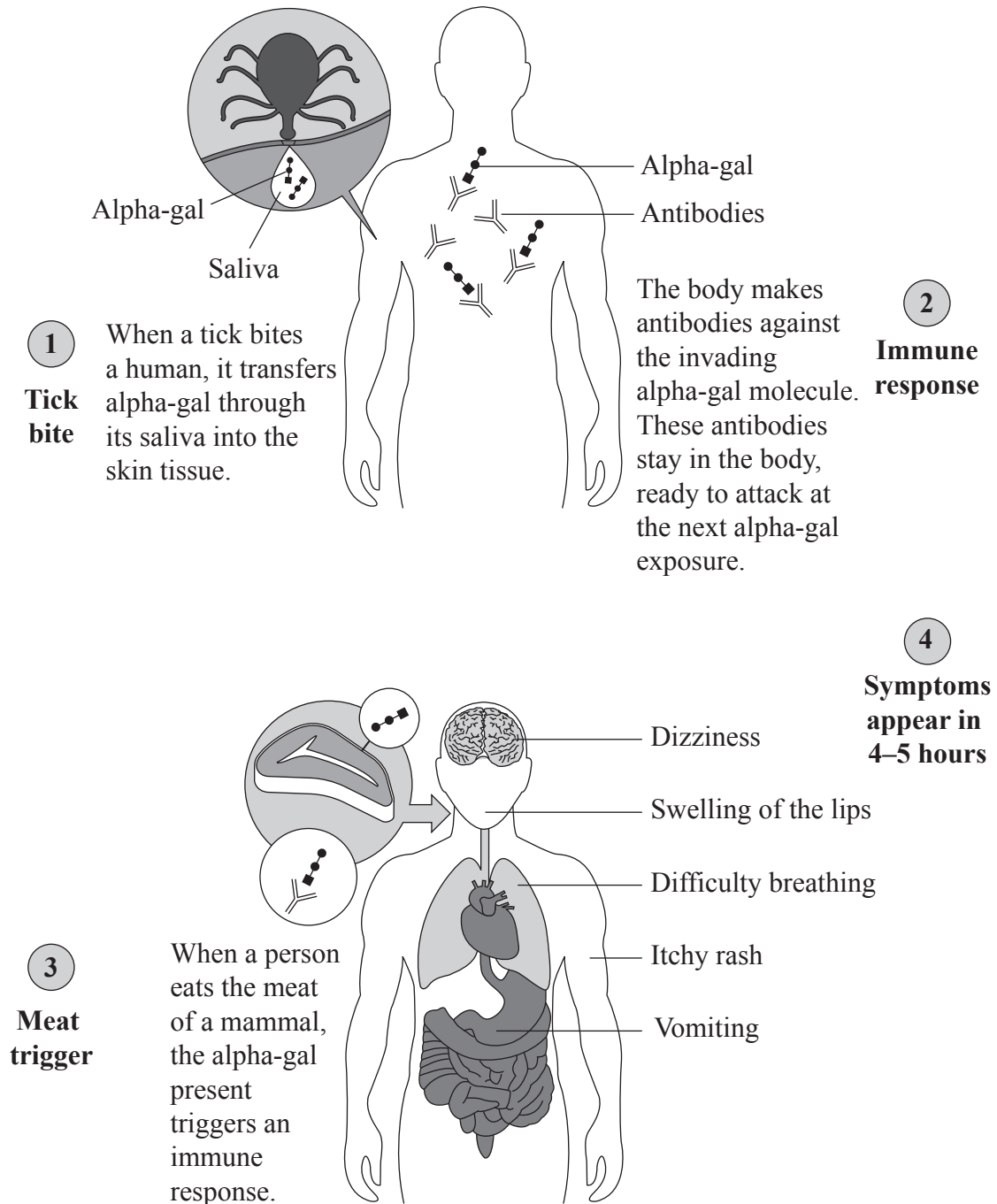
<i>Group A</i>		<i>Group B</i>	
<i>Animal</i>	<i>Number of eggs</i>	<i>Animal</i>	<i>Number of eggs</i>
Monotremes	1–3	Crabs	1000–2000
Snakes	1–100	Sea urchins	100 000 to 2 million
Birds	1–17	Squid	2000–3000

Compare the types of fertilisation that occur in group *A* and group *B* animals with reference to the data provided.

[illegible]

Question 28 (9 marks)

Alpha-gal syndrome (AGS) is a tick-borne allergy to red meat caused by tick bites. Alpha-gal is a sugar molecule found in most mammals but not humans, and can also be found in the saliva of ticks. The diagram shows how a tick bite might cause a person to develop an allergic reaction to red meat.



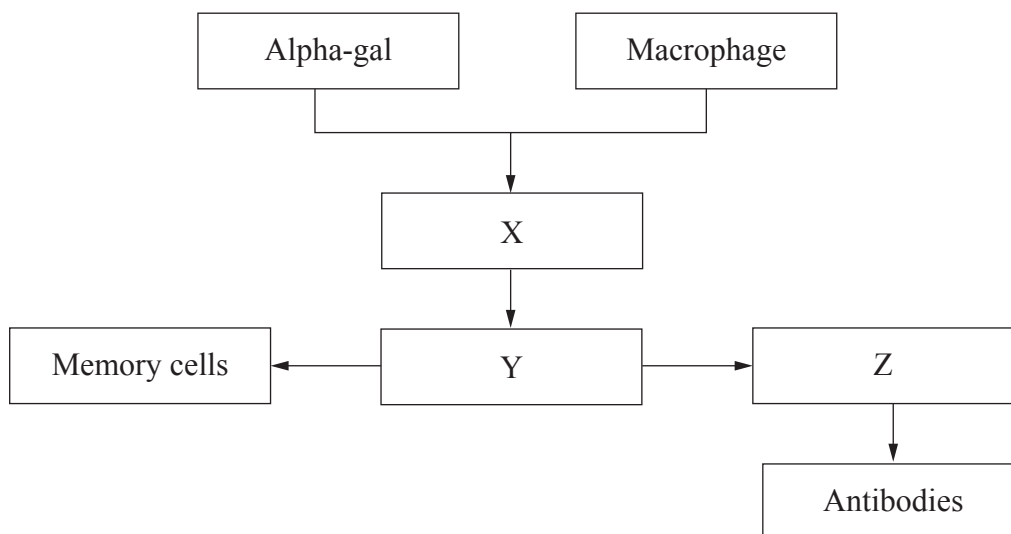
Courtesy Foothills Forum and Rappahannock News

Question 28 continues on page 21

Question 28 (continued)

- (a) The flow chart shows the process of antibody production following exposure to alpha-gal.

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Describe the role of X, Y and Z in the process of antibody production.

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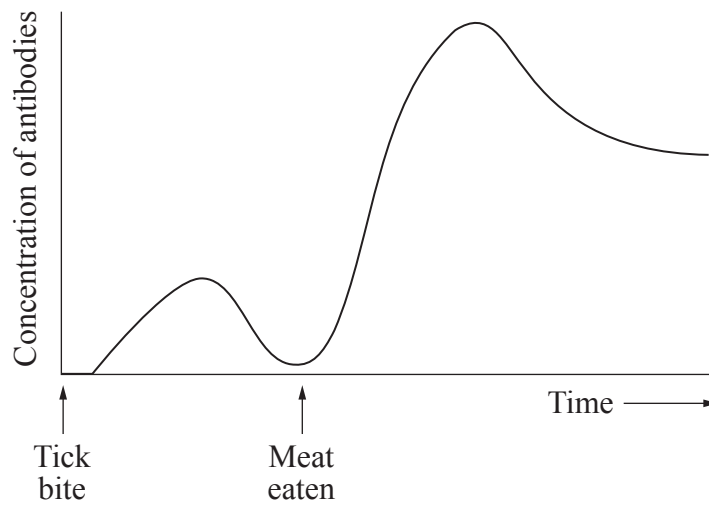
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Question 28 continues on page 22

Question 28 (continued)

- (b) An allergic reaction to alpha-gal sugar is similar to a secondary immune response.

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Describe the features of antibody production shown in the graph.

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- (c) Explain the role of memory cells in the immune response.

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End of Question 28

Question 29 (6 marks)

The *Varroa* mite is an external parasite of European honey bees and is considered to be the most serious pest of honey bees worldwide.

- (a) Why is *Varroa* mite infection considered to be an infectious disease? 2

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- (b) In June 2022, the *Varroa* mite was detected for the first time in Australia at the Port of Newcastle. It then spread to surrounding areas. 4

Explain TWO procedures that could have been employed to prevent the spread of the *Varroa* mite in honey bees.

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Question 30 (11 marks)

PAI-1 protein is encoded by the SERPINE 1 gene in humans. *Anopheles* mosquitoes have been genetically modified to express PAI-1, which blocks the entry of the malarial *Plasmodium* into the mosquito gut. This disrupts the *Plasmodium* life cycle, resulting in reduced transmission of malaria.

- (a) Describe a process that could be used to produce mosquitoes which express PAI-1.

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Question 30 continues on page 25

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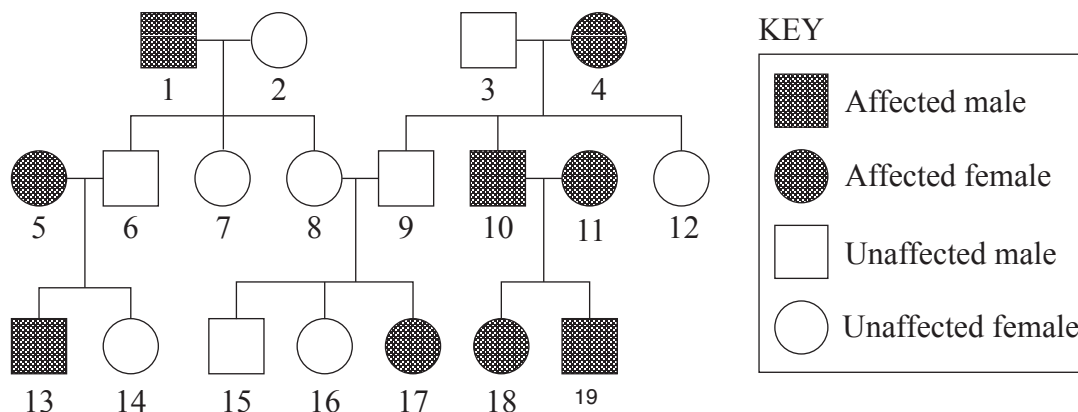
- 7

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– 25 –

Question 31 (6 marks)

Congenital amegakaryocytic thrombocytopenia (CAMT) is a rare, inherited disorder where bone marrow no longer makes platelets that are important for clotting and preventing bleeding. The pedigree below shows the inheritance of CAMT in a family.



Courtesy of Michigan Department of Health and Human Services (MDHHS)

- (a) What type of inheritance is shown in the pedigree above? Justify your answer.

3

Type of inheritance:

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Question 31 continues on page 27

Question 31 (continued)

- (b) A CAMT mutation was found to produce the following amino acid sequence.

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Glutamine – Tyrosine – Isoleucine – Aspartic acid

The same DNA fragment has been sequenced from an unaffected individual.

Template strand GTC ATA CAG CTG

The following codon chart displays all the codons and corresponding amino acids. The chart translates mRNA sequences into amino acids.

		Second base in codon					
		U	C	A	G		
First base in codon	U	UUU] Phenylalanine	UCU]	UAU] Tyrosine	UGU] Cysteine	U	Third base in codon
		UUC]	UCC] Serine	UAC]	UGC]	C	
		UUA] Leucine	UCA]	UAA] STOP	UGA – STOP	A	
		UUG]	UCG]	UAG]	UGG – Tryptophan	G	
	C	CUU]	CCU]	CAU] Histidine	CGU]	U	
		CUC] Leucine	CCC]	CAC]	CGC] Arginine	C	
		CUA]	CCA]	CAA] Glutamine	CGA]	A	
		CUG]	CCG]	CAG]	CGG]	G	
	A	AUU]	ACU]	AAU] Asparagine	AGU] Serine	U	
		AUC] Isoleucine	ACC]	AAC]	AGC]	C	
		AUA]	ACA]	AAA] Lysine	AGA] Arginine	A	
		AUG – Methionine (start)	ACG]	AAG]	AGG]	G	
	G	GUU]	GCU]	GAU] Aspartic acid	GGU]	U	
		GUC] Valine	GCC]	GAC]	GGC]	C	
		GUA]	GCA]	GAA] Glutamic acid	GGA]	A	
		GUG]	GCG]	GAG]	GGG]	G	

© Open Oregon Educational Resources

Use the codon chart shown to explain the type of mutation which causes CAMT.

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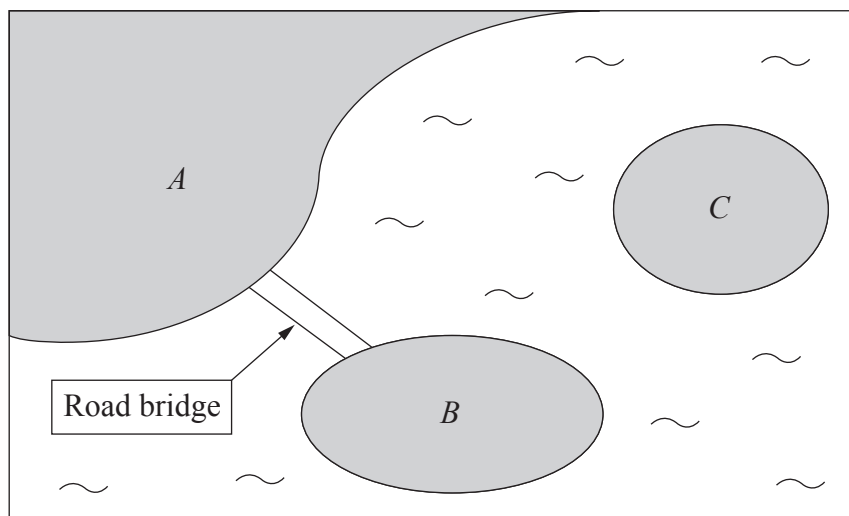
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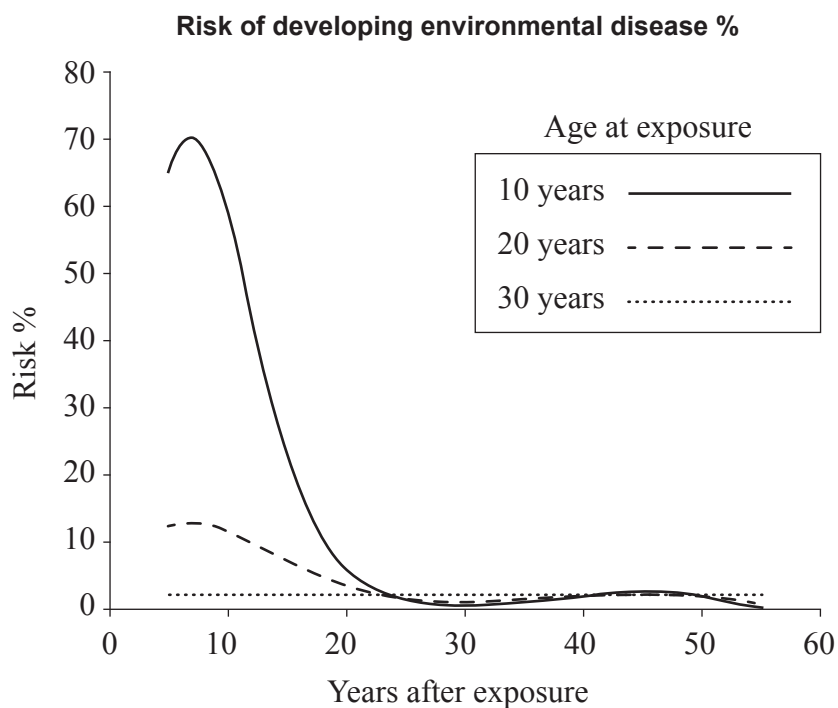
End of Question 31

Question 32 (7 marks)

A population lives across three regions, *A*, *B* and *C* as shown.



People in community *B* developed an environmental disease. An epidemiological study was carried out to determine the risk of developing the disease due to age at exposure. The results of this study are shown in the graph.



© National Academies Press

Question 32 continues on page 29

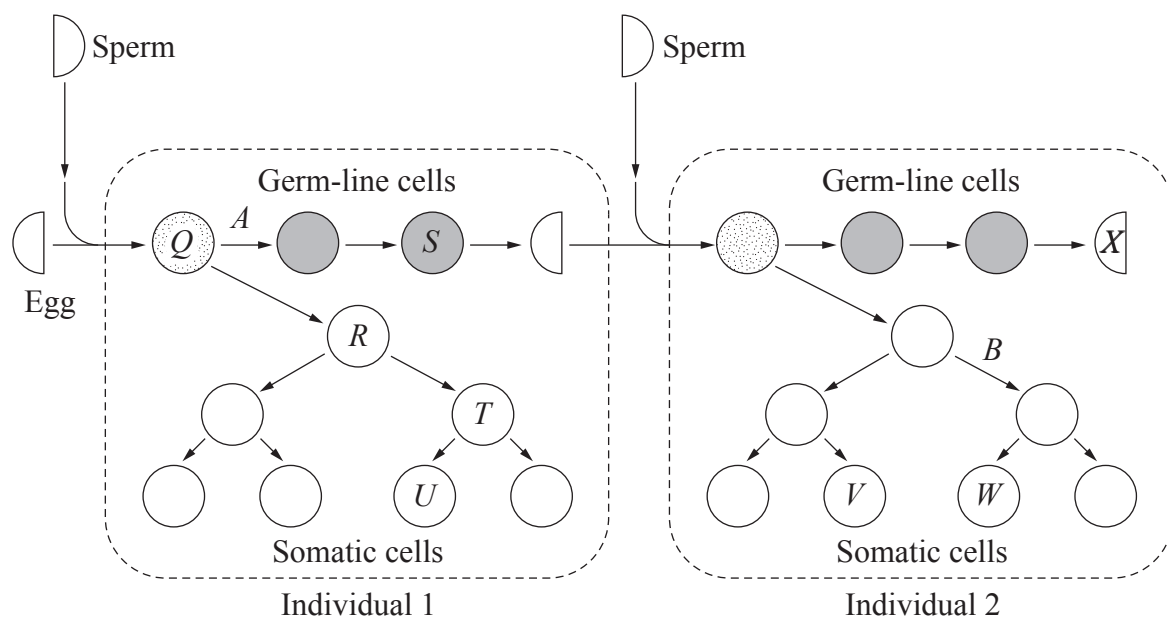
Design an epidemiological study that could be used to produce the results shown in the graph. Justify the features of your design.

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Question 33 (9 marks)

The following diagram shows the cell division processes occurring in two related individuals.



KEY

 or 	Gamete		Zygote		Germ-line cell		Somatic cell
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'Figure 19-5', from *ESSENTIAL CELL BIOLOGY*, THIRD EDITION by Bruce Alberts, et al.
 © 2004, 2010 by Bruce Alberts, Dennis Bray, Karen Hopkin, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, and Peter Walter. This selection may not be reproduced, stored in a retrieval system, or transmitted in any form or by any means without the prior written permission of the publisher.

- (a) Compare the cell division processes carried out by cells *R* and *S* in Individual 1.

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Question 33 continues on page 31

Question 33 (continued)

- (b) Explain the relationship between Individuals 1 and 2. 2

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- (c) *A* and *B* are two separate mutations. Analyse how mutations *A* and *B* affect the genetic information present in cells *U*, *V*, *W* and *X*. 4

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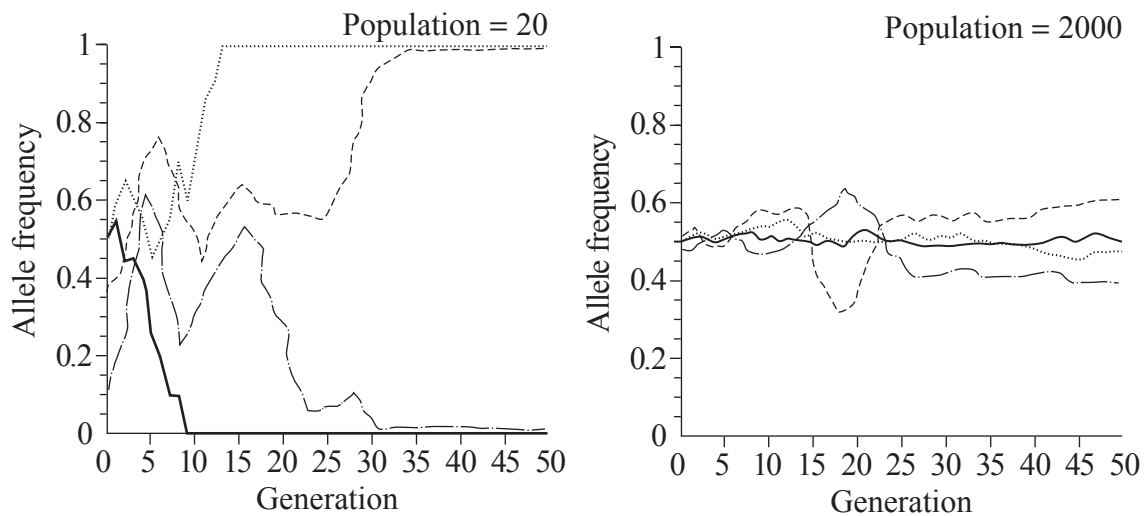
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End of Question 33

Question 34 (6 marks)

The following graphs show the changes in allele frequencies in two separate populations of the same species. Each line represents an introduced allele.



- (a) Explain why the fluctuations of the allele frequencies are more pronounced in the small population, compared to the larger population.

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Question 34 continues on page 33

Question 34 (continued)

- (b) Evaluate the effects of gene flow on the gene pools of the two populations.

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Section II extra writing space

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Handwriting practice lines for Section II extra writing space.

Section II extra writing space

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